Exercise 4 — Step-by-Step Solutions

Advanced System Theory · Discrete-Time Systems and Discretization

Course: Advanced System Theory (Paderborn University, Automatic Control Group RAT, Prof. Dr. Erdal Kayacan) **Covers:** Task 1 (a–c), Task 2 (a–d), Task 3, Task 4 (a–b), Task 5 (a–b) **Checking:** no official solution was available when this was written. Every result is verified by the scripts in `python/` — symbolic algebra (SymPy), step-by-step simulation of the difference equations, and comparison with `scipy.signal.cont2discrete`.

Every step is written out. References like "Concepts §6.3" point to `concepts.md`. Notation as in the lecture: $\boldsymbol{x}[k]$, $\boldsymbol{\Phi}[k,k_0]=\boldsymbol{A}^{k-k_0}$, $\boldsymbol{H}[k]$ impulse response, $\sigma[k]$ unit step, $\delta[k]$ unit impulse, T_s sampling interval, $\boldsymbol{A}_d$, $\boldsymbol{B}_d$ exact and $\tilde{\boldsymbol{A}}_d$, $\tilde{\boldsymbol{B}}_d$ forward-Euler discrete matrices.

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Task 1 — Response of discrete systems

$$\boldsymbol{x}[k+1]=\underbrace{\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}}_{\boldsymbol{A}}\boldsymbol{x}[k]+\underbrace{\begin{bmatrix}1\\ 1\end{bmatrix}}_{\boldsymbol{B}}u[k],\qquad y[k]=\underbrace{\begin{bmatrix}1 & 0\end{bmatrix}}_{\boldsymbol{C}}\boldsymbol{x}[k],\qquad \boldsymbol{x}[0]=\begin{bmatrix}1\\ -1\end{bmatrix},\qquad u[k]=\sigma[k].$$

The step means $u[k]=1$ for every $k\geq 0$. Note $\boldsymbol{A}\begin{bmatrix}a\\ b\end{bmatrix}=\begin{bmatrix}b\\ -a\end{bmatrix}$ (swap, then negate the second entry).

1(a) — First three steps

Run the recipe $\boldsymbol{x}[k+1]=\boldsymbol{A}\boldsymbol{x}[k]+\boldsymbol{B}\cdot 1$ (Concepts §2.2):

$$\boldsymbol{x}[1]=\begin{bmatrix}0\cdot 1+1\cdot(-1)\\ -1\cdot 1+0\cdot(-1)\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}=\begin{bmatrix}-1\\ -1\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}=\boxed{\begin{bmatrix}0\\ 0\end{bmatrix}}$$

$$\boldsymbol{x}[2]=\boldsymbol{A}\begin{bmatrix}0\\ 0\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}=\boxed{\begin{bmatrix}1\\ 1\end{bmatrix}},\qquad \boldsymbol{x}[3]=\begin{bmatrix}1\\ -1\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}=\boxed{\begin{bmatrix}2\\ 0\end{bmatrix}}$$

One more step is revealing: $\boldsymbol{x}[4] = \begin{bmatrix} 0 \\ -2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \boldsymbol{x}[0]$. **The state returns to its start after 4 steps** — the response is periodic.

1(b) — General solution $x[k]$

Step 1 — Powers of $\boldsymbol{A}$ (Concepts §4.3). $\boldsymbol{A}^2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -\boldsymbol{I}$, hence $\boldsymbol{A}^3 = -\boldsymbol{A}$, $\boldsymbol{A}^4 = \boldsymbol{I}$, and

$$\boldsymbol{A}^k = \cos\left(\frac{k\pi}{2}\right)\boldsymbol{I} + \sin\left(\frac{k\pi}{2}\right)\boldsymbol{A} = \begin{bmatrix} \cos\frac{k\pi}{2} & \sin\frac{k\pi}{2} \\ -\sin\frac{k\pi}{2} & \cos\frac{k\pi}{2} \end{bmatrix}.$$

Check: $k = 0, 1, 2, 3$ gives $\boldsymbol{I}, \boldsymbol{A}, -\boldsymbol{I}, -\boldsymbol{A}$ ✓. (Eigenvalues $\pm j = e^{\pm j\pi/2}$: $|\lambda| = 1$, a quarter turn per step.)

Step 2 — General formula (Concepts §5.1):

$$\boldsymbol{x}[k] = \boldsymbol{A}^k\boldsymbol{x}[0] + \sum_{j=0}^{k-1} \boldsymbol{A}^{(k-1)-j}\boldsymbol{B}u[j] = \boldsymbol{A}^k\boldsymbol{x}[0] + \sum_{i=0}^{k-1} \boldsymbol{A}^i\boldsymbol{B}.$$

Step 3 — Evaluate the sum with the fixed point (Concepts §6.3). 1 is not an eigenvalue of $\boldsymbol{A}$, so the constant input has a fixed point $\boldsymbol{x}^* = \boldsymbol{A}\boldsymbol{x}^* + \boldsymbol{B}$:

$$\boldsymbol{I} - \boldsymbol{A} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad (\boldsymbol{I} - \boldsymbol{A})^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \boldsymbol{x}^* = (\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} = \frac{1}{2}\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Check: $\boldsymbol{A}\boldsymbol{x}^* + \boldsymbol{B} = [0, -1]^\top + [1, 1]^\top = [1, 0]^\top$ ✓. Then $\boldsymbol{x}[k] = \boldsymbol{A}^k(\boldsymbol{x}[0] - \boldsymbol{x}^*) + \boldsymbol{x}^*$ with $\boldsymbol{x}[0] - \boldsymbol{x}^* = [0, -1]^\top$:

$$\boldsymbol{A}^k \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -\sin\frac{k\pi}{2} \\ -\cos\frac{k\pi}{2} \end{bmatrix} \quad \Longrightarrow \quad \boxed{\boldsymbol{x}[k] = \begin{bmatrix} 1 - \sin\frac{k\pi}{2} \\ -\cos\frac{k\pi}{2} \end{bmatrix}, \quad k \geq 0}$$

(Equivalently, without the fixed point: $\sum_{i=0}^{k-1} \boldsymbol{A}^i\boldsymbol{B} = (\boldsymbol{I} - \boldsymbol{A})^{-1}(\boldsymbol{I} - \boldsymbol{A}^k)\boldsymbol{B}$ — same result.)

Check against (a): $k = 0$: $[1, -1]^\top$ ✓; $k = 1$: $[1-1, -0]^\top = [0, 0]^\top$ ✓; $k = 2$: $[1-0, -(-1)]^\top = [1, 1]^\top$ ✓; $k = 3$: $[1+1, -0]^\top = [2, 0]^\top$ ✓.

Interpretation. The state **circles around the fixed point** $\boldsymbol{x}^* = [1, 0]^\top$ at distance 1, making a quarter turn per step: $[1, -1] \to [0, 0] \to [1, 1] \to [2, 0] \to [1, -1] \to \ldots$ Because $|\lambda| = 1$ the deviation from $\boldsymbol{x}^*$ neither decays nor grows — the step response **never settles** (marginally stable).

1(c) — Total output response

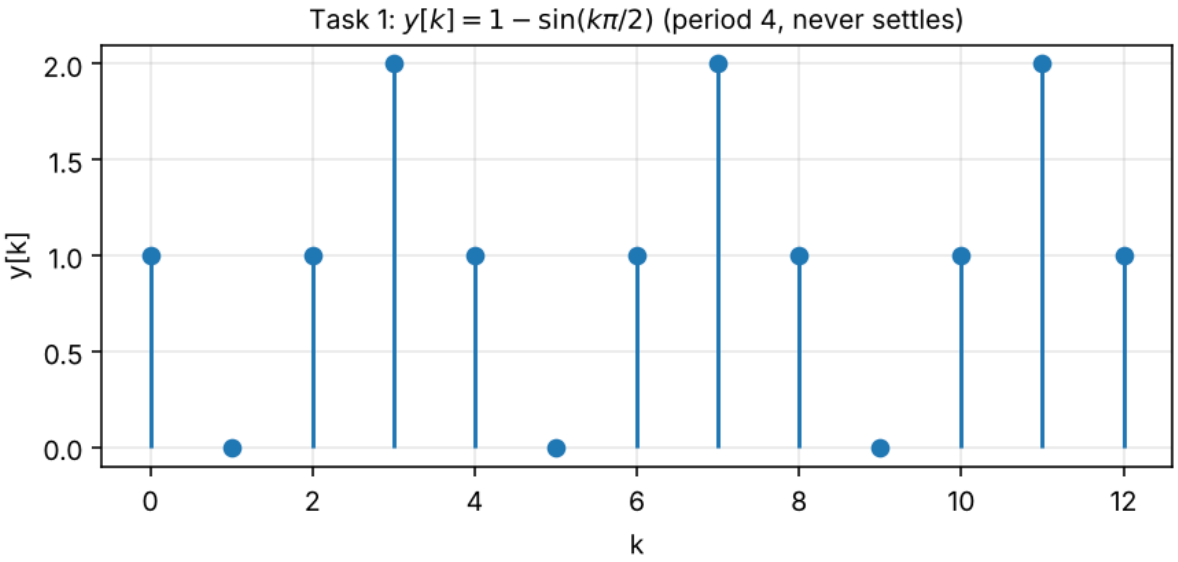
$\boldsymbol{D} = 0$, so $y[k] = \boldsymbol{C}\boldsymbol{x}[k] = x_1[k]$:

$$\boxed{y[k] = 1 - \sin\left(\frac{k\pi}{2}\right), \quad k \geq 0 \qquad \Longrightarrow \qquad y = 1,\ 0,\ 1,\ 2,\ 1,\ 0,\ 1,\ 2,\ \ldots}$$

Split into the two parts (Concepts §5.2):

- zero-input: $\boldsymbol{C}\boldsymbol{A}^k\boldsymbol{x}[0] = \cos\frac{k\pi}{2} - \sin\frac{k\pi}{2}$ (the first row of $\boldsymbol{A}^k$ times $[1,-1]^\top$);
- zero-state: $\boldsymbol{C}(\boldsymbol{I}-\boldsymbol{A})^{-1}(\boldsymbol{I}-\boldsymbol{A}^k)\boldsymbol{B} = 1 - \cos\frac{k\pi}{2}$;
- sum: $1 - \sin\frac{k\pi}{2}$ ✓.

Check: `python/task1_discrete_response.py` (exact rational arithmetic for $k = 0, \ldots, 12$).



Task 2 — Reachable states

$$\boldsymbol{x}[k+1] = \underbrace{\begin{bmatrix} 0 & 1 \\ -0.5 & -1 \end{bmatrix}}_{\boldsymbol{A}} \boldsymbol{x}[k] + \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\boldsymbol{B}} u[k].$$

2(a) — From $\boldsymbol{x}[0] = 0$, at most 2 steps are needed

Step 1 — State after k steps. With $\boldsymbol{x}[0] = 0$ (Concepts §7.2):

$$\boldsymbol{x}[k] = \sum_{j=0}^{k-1} \boldsymbol{A}^{(k-1)-j}\boldsymbol{B}u[j] = \boldsymbol{B}u[k-1] + \boldsymbol{A}\boldsymbol{B}u[k-2] + \boldsymbol{A}^2\boldsymbol{B}u[k-3] + \cdots + \boldsymbol{A}^{k-1}\boldsymbol{B}u[0].$$

Every reachable state is a linear combination of $\boldsymbol{B}, \boldsymbol{A}\boldsymbol{B}, \boldsymbol{A}^2\boldsymbol{B}, \ldots$

Step 2 — Cayley-Hamilton. $\det(\lambda\boldsymbol{I} - \boldsymbol{A}) = \lambda(\lambda+1) + 0.5 = \lambda^2 + \lambda + 0.5$, so

$$\boldsymbol{A}^2 + \boldsymbol{A} + 0.5\boldsymbol{I} = 0 \quad \Longrightarrow \quad \boldsymbol{A}^2 = -\boldsymbol{A} - 0.5\boldsymbol{I}.$$

Check: $\boldsymbol{A}^2 = \begin{bmatrix} -0.5 & -1 \\ 0.5 & 0.5 \end{bmatrix}$ and $-\boldsymbol{A} - 0.5\boldsymbol{I} = \begin{bmatrix} -0.5 & -1 \\ 0.5 & 1-0.5 \end{bmatrix}$ ✓.

Step 3 — Higher powers add nothing new. Multiply by $\boldsymbol{B}$: $\boldsymbol{A}^2\boldsymbol{B} = -\boldsymbol{A}\boldsymbol{B} - 0.5\boldsymbol{B} \in \text{span}\{\boldsymbol{B}, \boldsymbol{A}\boldsymbol{B}\}$. Multiplying the identity by $\boldsymbol{A}$ again, $\boldsymbol{A}^3\boldsymbol{B} = -\boldsymbol{A}^2\boldsymbol{B} - 0.5\boldsymbol{A}\boldsymbol{B}$ is also in that span, and by induction every $\boldsymbol{A}^m\boldsymbol{B}$, $m \geq 2$. Therefore, for any $k \geq 2$, $\boldsymbol{x}[k]$ can be rewritten as

$$\boldsymbol{x}[k] = \boldsymbol{B}\, v_1 + \boldsymbol{A}\boldsymbol{B}\, v_0 \quad \text{for some numbers } v_0, v_1,$$

which is exactly what **two** steps produce with $u[0] = v_0$, $u[1] = v_1$: $\boldsymbol{x}[2] = \boldsymbol{A}\boldsymbol{B}u[0] + \boldsymbol{B}u[1]$.

Step 4 — Conclusion. Any state that can be reached in some number k of steps lies in $\text{span}\{\boldsymbol{B}, \boldsymbol{A}\boldsymbol{B}\}$ and can therefore be reached in 2 steps. (If $k = 1$ it is reachable in one step, which is "at most 2".) ∎

Here every state is reachable: $\boldsymbol{A}\boldsymbol{B} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 2 \\ -0.5 \cdot 1 - 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2.5 \end{bmatrix}$ and

$$\det\begin{bmatrix} \boldsymbol{B} & \boldsymbol{A}\boldsymbol{B} \end{bmatrix} = \det\begin{bmatrix} 1 & 2 \\ 2 & -2.5 \end{bmatrix} = -2.5 - 4 = -6.5 \neq 0,$$

so $\boldsymbol{B}$ and $\boldsymbol{A}\boldsymbol{B}$ span all of $\mathbb{R}^2$.

2(b) — Inputs that reach $\boldsymbol{x}_{des} = [4,\ 2]^\top$

One step is not enough: $\boldsymbol{x}[1] = \boldsymbol{B}u[0] = [u[0],\ 2u[0]]^\top$ always has $x_2 = 2x_1$, but $\boldsymbol{x}_{des}$ has $x_2 = 2 \neq 2 \cdot 4 = 8$. So two steps are needed.

Two steps: $\boldsymbol{x}[2] = \boldsymbol{A}\boldsymbol{B}u[0] + \boldsymbol{B}u[1]$:

$$u[0]\begin{bmatrix} 2 \\ -2.5 \end{bmatrix} + u[1]\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \iff \begin{cases} 2u[0] + u[1] = 4 \\ -2.5u[0] + 2u[1] = 2 \end{cases}$$

From the first equation $u[1] = 4 - 2u[0]$; insert into the second: $-2.5u[0] + 8 - 4u[0] = 2 \Rightarrow 6.5u[0] = 6$:

$$\boxed{u[0] = \frac{12}{13} \approx 0.923, \qquad u[1] = 4 - \frac{24}{13} = \frac{28}{13} \approx 2.154}$$

Check by simulation:

$$\boldsymbol{x}[1] = \boldsymbol{B}u[0] = \frac{1}{13}\begin{bmatrix} 12 \\ 24 \end{bmatrix}, \qquad \boldsymbol{x}[2] = \boldsymbol{A}\boldsymbol{x}[1] + \boldsymbol{B}u[1] = \frac{1}{13}\begin{bmatrix} 24 \\ -6 - 24 \end{bmatrix} + \frac{1}{13}\begin{bmatrix} 28 \\ 56 \end{bmatrix} = \frac{1}{13}\begin{bmatrix} 52 \\ 26 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \checkmark$$

The two-step solution is **unique** (det $\neq 0$). With more steps there are infinitely many input sequences (more unknowns than equations).

2(c) — General $\boldsymbol{A} \in \mathbb{R}^{n \times n}$: at most n steps

Claim: with $\boldsymbol{x}[0] = 0$, every state that is reachable at all is reachable in at most n steps.

Proof (Concepts §7.3).

1. *Reachable set after k steps.* $\boldsymbol{x}[k] = \sum_{i=0}^{k-1} \boldsymbol{A}^i \boldsymbol{B} u[k-1-i] = \boldsymbol{R}_k \boldsymbol{U}_k$ with $\boldsymbol{R}_k = [\boldsymbol{B} \;\; \boldsymbol{AB} \;\; \cdots \;\; \boldsymbol{A}^{k-1}\boldsymbol{B}]$, so the set $\mathcal{R}_k$ of states reachable in exactly k steps is $\operatorname{Im} \boldsymbol{R}_k$ (all combinations of its columns). This also holds for several inputs ($\boldsymbol{B} \in \mathbb{R}^{n\times m}$).
2. *Cayley–Hamilton.* With $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$: $\boldsymbol{A}^n = -c_{n-1}\boldsymbol{A}^{n-1} - \cdots - c_1 \boldsymbol{A} - c_0 \boldsymbol{I}$.
3. *Induction.* For every $m \geq n$, $\boldsymbol{A}^m \boldsymbol{B}$ is a linear combination of $\boldsymbol{B}, \ldots, \boldsymbol{A}^{n-1}\boldsymbol{B}$. Base $m = n$: multiply step 2 by $\boldsymbol{B}$. Step $m \to m+1$: $\boldsymbol{A}^{m+1}\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{A}^m \boldsymbol{B}) = \sum_{i=0}^{n-1} \beta_i \boldsymbol{A}^{i+1}\boldsymbol{B}$; every term except $\beta_{n-1}\boldsymbol{A}^n\boldsymbol{B}$ is already in the list, and that one is by the base case.
4. *Consequence.* For $k \geq n$ the extra columns of $\boldsymbol{R}_k$ are combinations of the first n blocks, so $\mathcal{R}_k = \operatorname{Im} \boldsymbol{R}_k = \operatorname{Im} \boldsymbol{R}_n = \mathcal{R}_n$.
5. *Fewer steps are covered too.* For $k < n$: $\mathcal{R}_k \subseteq \mathcal{R}_n$, because the inputs $u[0] = \cdots = u[n-k-1] = 0$ keep the state at 0 and the remaining k inputs reproduce the k-step result. (Equivalently: the columns of $\boldsymbol{R}_k$ are among those of $\boldsymbol{R}_n$.)
6. Hence $\bigcup_k \mathcal{R}_k = \mathcal{R}_n$: anything reachable is reachable in n steps. ∎

The matrix $\boldsymbol{R}_n$ is the **reachability matrix**; if $\operatorname{rank} \boldsymbol{R}_n = n$, every state is reachable (Week 11).

2(d) — (Hard) Nonzero initial state

No — the statement does not hold for an arbitrary system with $\boldsymbol{x}[0] \neq 0$.

Why the proof breaks. Now $\boldsymbol{x}[k] = \boldsymbol{A}^k \boldsymbol{x}[0] + \boldsymbol{R}_k \boldsymbol{U}_k$. The states reachable at step k form the shifted set $\boldsymbol{A}^k \boldsymbol{x}[0] + \operatorname{Im} \boldsymbol{R}_k$. Cayley–Hamilton still freezes $\operatorname{Im} \boldsymbol{R}_k$ for $k \geq n$, but the **shift** $\boldsymbol{A}^k \boldsymbol{x}[0]$ **keeps moving** with k. Step 5 of the proof also fails: a zero input no longer keeps the state still, so "waiting" changes where you are.

Counterexample ($n = 2$):

$$\boldsymbol{A} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \boldsymbol{x}[0] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The input only affects x_2, and x_1 evolves freely: $x_1[k] = 2^k$. The target $\boldsymbol{x}_{des} = [8,\ 0]^\top$ is reached at $k = 3$ (choose $u[0] = u[1] = u[2] = 0$), but **at no** $k \leq 2$, since $x_1[1] = 2$ and $x_1[2] = 4$. So a state is reachable, yet only after $3 > n$ steps. (Similarly $[2^m, 0]^\top$ needs exactly m steps, for any m.)

When does it still hold? If the system is **reachable**, $\operatorname{rank} \boldsymbol{R}_n = n$, then for every $\boldsymbol{x}[0]$ and every $\boldsymbol{x}_{des}$ the equation $\boldsymbol{R}_n \boldsymbol{U}_n = \boldsymbol{x}_{des} - \boldsymbol{A}^n \boldsymbol{x}[0]$ has a solution, so n steps always suffice. The system of 2(a) is of this kind ($\det[\boldsymbol{B}\ \boldsymbol{AB}] = -6.5 \neq 0$), so for it the statement *does* hold from any initial state. The counterexample has $\operatorname{rank}[\boldsymbol{B}\ \boldsymbol{AB}] = \operatorname{rank} \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = 1 < 2$.

Check: `python/task2_reachability.py`.

Task 3 — Response of discrete systems 2

Two systems, both with $\boldsymbol{x}[0] = [1,\ -1]^\top$:

System 1: $\boldsymbol{A} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, $\boldsymbol{B} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\boldsymbol{C} = [1 \quad 1]$; System 2: $\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$, $\boldsymbol{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\boldsymbol{C} = [1 \quad 0$

$\boldsymbol{D} = 0$ in both. Plan for each system (Concepts §5.2, §6):

$$y[k] = \underbrace{\boldsymbol{C}\boldsymbol{A}^k\boldsymbol{x}[0]}_{y_{zi}[k]} + \underbrace{\sum_{j=0}^{k-1} \boldsymbol{C}\boldsymbol{A}^{(k-1)-j}\boldsymbol{B}u[j]}_{y_{zs}[k]}, \qquad y_{zs}^{\delta}[k] = H[k] = \boldsymbol{C}\boldsymbol{A}^{k-1}\boldsymbol{B}, \qquad y_{zs}^{\sigma}[k] = \sum_{i=0}^{k-1} \boldsymbol{C}\boldsymbol{A}^{i}\boldsymbol{B} \quad (k \geq 1$$

At $k = 0$ the zero-state part is $\boldsymbol{D}u[0] = 0$, so $y[0] = \boldsymbol{C}\boldsymbol{x}[0]$ for both inputs.

System 1 (diagonal)

Powers. $\boldsymbol{A}$ is diagonal: $\boldsymbol{A}^k = \text{diag}\left((-1)^k, (-2)^k\right)$.

Zero-input response.

$$\boldsymbol{A}^k\boldsymbol{x}[0] = \begin{bmatrix} (-1)^k \\ -(-2)^k \end{bmatrix}, \qquad y_{zi}[k] = (-1)^k - (-2)^k.$$

Impulse response ($k \geq 1$):

$$\boldsymbol{A}^{k-1}\boldsymbol{B} = \begin{bmatrix} (-1)^{k-1} \\ -(-2)^{k-1} \end{bmatrix}, \qquad H[k] = (-1)^{k-1} - (-2)^{k-1}, \qquad H[0] = 0.$$

Total response to $u = \delta[k]$. $k = 0$: $y[0] = 1 - 1 = 0$. For $k \geq 1$:

$$y[k] = (-1)^k - (-2)^k + (-1)^{k-1} - (-2)^{k-1} = \underbrace{(-1)^{k-1}(-1+1)}_{0} - (-2)^{k-1}(-2+1) = (-2)^{k-1}.$$

System 1, impulse: $y[0] = 0, \qquad y[k] = (-2)^{k-1}\ (k \geq 1): \quad 0,\ 1,\ -2,\ 4,\ -8,\ \ldots$

Step response. With the scalar geometric sums $\sum_{i=0}^{k-1}(-1)^i = \frac{1-(-1)^k}{2}$ and $\sum_{i=0}^{k-1}(-2)^i = \frac{1-(-2)^k}{3}$ (Concepts §6.3):

$$y_{zs}[k] = \sum_{i=0}^{k-1}\left[(-1)^i - (-2)^i\right] = \frac{1-(-1)^k}{2} - \frac{1-(-2)^k}{3} = \frac{1}{6} - \frac{1}{2}(-1)^k + \frac{1}{3}(-2)^k.$$

Adding y_{zi}:

System 1, step: $y[k] = \dfrac{1}{6} + \dfrac{1}{2}(-1)^k - \dfrac{2}{3}(-2)^k, \quad k \geq 0: \quad 0,\ 1,\ -2,\ 5,\ -10,\ \ldots$

(The formula also gives the right value 0 at $k = 0$.) *Check by iteration:* $\boldsymbol{x}[1] = \boldsymbol{A}\boldsymbol{x}[0] + \boldsymbol{B} = [-1, 2]^\top + [1, -1]^\top = [0, 1]^\top$, $y[1] = 1$ ✓; $\boldsymbol{x}[2] = [0, -2]^\top + [1, -1]^\top = [1, -3]^\top$, $y[2] = -2$ ✓ (formula: $\frac{1}{6} + \frac{1}{2} - \frac{8}{3} = -2$ ✓).

System 2 (companion form)

Step 1 — Eigenvalues and eigenvectors. $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \lambda(\lambda+3)+2 = (\lambda+1)(\lambda+2) \Rightarrow \lambda_1 = -1$, $\lambda_2 = -2$ (same as system 1).

- $\lambda_1 = -1$: $\lambda_1 \boldsymbol{I} - \boldsymbol{A} = \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix} \Rightarrow p_1 + p_2 = 0 \Rightarrow \boldsymbol{p}_1 = [1,\ -1]^\top$.
- $\lambda_2 = -2$: $\lambda_2 \boldsymbol{I} - \boldsymbol{A} = \begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix} \Rightarrow 2p_1 + p_2 = 0 \Rightarrow \boldsymbol{p}_2 = [1,\ -2]^\top$.

Step 2 — Decompose the vectors we need to propagate (Concepts §4.1, "decompose the vector"):

- $\boldsymbol{x}[0] = [1,-1]^\top = \boldsymbol{p}_1$ **— already an eigenvector**, so $\boldsymbol{A}^k \boldsymbol{x}[0] = (-1)^k \boldsymbol{p}_1$.
- $\boldsymbol{B} = [0,1]^\top = c_1 \boldsymbol{p}_1 + c_2 \boldsymbol{p}_2$: $c_1 + c_2 = 0$, $-c_1 - 2c_2 = 1 \Rightarrow c_2 = -1$, $c_1 = 1$. So $\boldsymbol{B} = \boldsymbol{p}_1 - \boldsymbol{p}_2$ and $\boldsymbol{A}^m \boldsymbol{B} = (-1)^m \boldsymbol{p}_1 - (-2)^m \boldsymbol{p}_2$.

Step 3 — Zero-input response and impulse response. $\boldsymbol{C} = [1\ 0]$ picks the first entry, which is 1 in both $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$:

$$y_{zi}[k] = (-1)^k, \qquad H[k] = \boldsymbol{C}\boldsymbol{A}^{k-1}\boldsymbol{B} = (-1)^{k-1} - (-2)^{k-1}\ (k \geq 1), \qquad H[0] = 0.$$

Step 4 — Total response to $u = \delta[k]$. $k = 0$: $y[0] = \boldsymbol{C}\boldsymbol{x}[0] = 1$. For $k \geq 1$: $y[k] = (-1)^k + (-1)^{k-1} - (-2)^{k-1} = -(-2)^{k-1}$.

System 2, impulse: $y[0] = 1, \qquad y[k] = -(-2)^{k-1}\ (k \geq 1): \quad 1,\ -1,\ 2,\ -4,\ 8,\ \ldots$

Step 5 — Step response. y_{zs} is the same geometric sum as for system 1, $\frac{1}{6} - \frac{1}{2}(-1)^k + \frac{1}{3}(-2)^k$; adding $y_{zi} = (-1)^k$:

System 2, step: $y[k] = \dfrac{1}{6} + \dfrac{1}{2}(-1)^k + \dfrac{1}{3}(-2)^k, \quad k \geq 0: \quad 1,\ -1,\ 2,\ -3,\ 6,\ \ldots$

Check by iteration: $\boldsymbol{x}[1] = \boldsymbol{A}\boldsymbol{x}[0] + \boldsymbol{B} = [-1,1]^\top + [0,1]^\top = [-1,2]^\top$, $y[1] = -1$ ✓; $\boldsymbol{x}[2] = [2, 2-6]^\top + [0,1]^\top = [2,-3]^\top$, $y[2] = 2$ ✓.

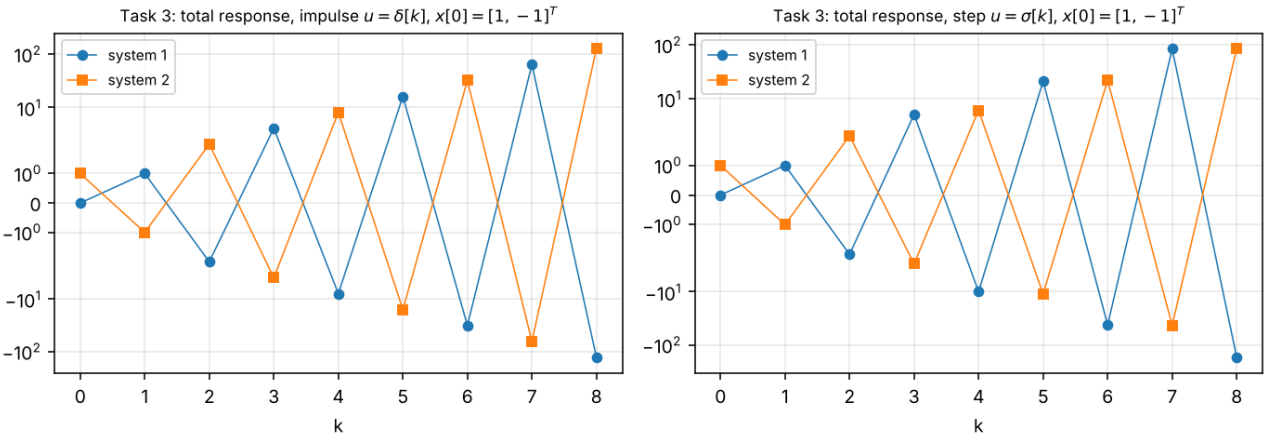
Observations

	System 1	System 2
eigenvalues	$-1, -2$	$-1, -2$
impulse response $H[k]$ $(k \geq 1)$	$(-1)^{k-1} - (-2)^{k-1}$	$(-1)^{k-1} - (-2)^{k-1}$
zero-input response	$(-1)^k - (-2)^k$	$(-1)^k$
impulse, total	$(-2)^{k-1}$	$-(-2)^{k-1}$
step, total	$\frac{1}{6} + \frac{1}{2}(-1)^k - \frac{2}{3}(-2)^k$	$\frac{1}{6} + \frac{1}{2}(-1)^k + \frac{1}{3}(-2)^k$

- **Same impulse response:** the two systems have identical **input–output** behaviour from $\boldsymbol{x}[0] = 0$ — they are two different state-space *realizations* of the same system (Weeks 6 and 13–14; both have transfer function $\frac{1}{(z+1)(z+2)}$). They differ only through the initial state: in system 2, $\boldsymbol{x}[0]$ excites only the $\lambda = -1$ mode.

- **Both responses diverge** with alternating sign: the eigenvalue -2 has $|\lambda| = 2 > 1$ (Concepts §4.4), and the input excites it. Both systems are unstable.

Check: `python/task3_discrete_responses.py` (exact arithmetic, $k = 0, \ldots, 11$).



Task 4 — Discretization

For each system: exact (ZOH) discretization $\boldsymbol{A}_d = e^{\boldsymbol{A}T_s}$, $\boldsymbol{B}_d = \int_0^{T_s} e^{\boldsymbol{A}\alpha}\mathrm{d}\alpha\, \boldsymbol{B}$ (Concepts §8), and forward Euler $\tilde{\boldsymbol{A}}_d = \boldsymbol{I} + T_s\boldsymbol{A}$, $\tilde{\boldsymbol{B}}_d = T_s\boldsymbol{B}$ (Concepts §9). $\boldsymbol{C}$ and $\boldsymbol{D}$ are unchanged. **Both $\boldsymbol{A}$ are singular**, so $\boldsymbol{B}_d = \boldsymbol{A}^{-1}(\boldsymbol{A}_d - \boldsymbol{I})\boldsymbol{B}$ cannot be used; the integral is computed directly (Concepts §8.4).

4(a) — Double integrator $\ddot{y}(t) = u(t)$

State-space model (Week 2 recipe, $x_1 = y$, $x_2 = \dot{y}$): $\dot{x}_1 = x_2$, $\dot{x}_2 = u$:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \boldsymbol{D} = 0.$$

Matrix exponential. $\boldsymbol{A}^2 = 0$ (nilpotent), so $e^{\boldsymbol{A}\alpha} = \boldsymbol{I} + \boldsymbol{A}\alpha = \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix}$.

Exact discretization.

$$\boldsymbol{A}_d = e^{\boldsymbol{A}T_s} = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{B}_d = \int_0^{T_s} \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mathrm{d}\alpha = \int_0^{T_s} \begin{bmatrix} \alpha \\ 1 \end{bmatrix} \mathrm{d}\alpha = \begin{bmatrix} T_s^2/2 \\ T_s \end{bmatrix}.$$

$$\boxed{\boldsymbol{x}[k+1] = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix} \boldsymbol{x}[k] + \begin{bmatrix} T_s^2/2 \\ T_s \end{bmatrix} u[k], \qquad y[k] = \begin{bmatrix} 1 & 0 \end{bmatrix} \boldsymbol{x}[k]}$$

Forward Euler.

$$\boxed{\tilde{\boldsymbol{A}}_d = \boldsymbol{I} + T_s\boldsymbol{A} = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}, \qquad \tilde{\boldsymbol{B}}_d = T_s\boldsymbol{B} = \begin{bmatrix} 0 \\ T_s \end{bmatrix}, \qquad \boldsymbol{C},\ \boldsymbol{D} \text{ unchanged}}$$

Comparison. $\tilde{\boldsymbol{A}}_d = \boldsymbol{A}_d$ exactly, because $\boldsymbol{A}$ is nilpotent and the exponential series stops after the linear term (Concepts §9.2). The input matrices differ in the first entry: exact discretization contains the school formula $\Delta y = \dot{y}\,T_s + \frac{1}{2}uT_s^2$ — the distance covered **during** the interval due to the acceleration u. Euler assumes the velocity is constant over the step and misses the $\frac{1}{2}uT_s^2$ term.

4(b) — LTI system

$$\dot{\boldsymbol{x}}(t) = \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} \boldsymbol{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t), \qquad y(t) = \begin{bmatrix} 0 & 1 \end{bmatrix} \boldsymbol{x}(t).$$

($\dot{x}_1 = -x_1 + u$ is a first-order lag; $\dot{x}_2 = x_1$ integrates it.)

Step 1 — Eigenvalues. Lower triangular ⇒ $\lambda_1 = -1$, $\lambda_2 = 0$ (det $\boldsymbol{A} = 0$, singular).

Step 2 — $e^{\boldsymbol{A}t}$ with Cayley–Hamilton (Week 3; 2×2 formula with $\lambda_1 = -1$, $\lambda_2 = 0$):

$$\alpha_1(t) = \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2} = \frac{e^{-t} - 1}{-1} = 1 - e^{-t}, \qquad \alpha_0(t) = \frac{\lambda_1 e^{\lambda_2 t} - \lambda_2 e^{\lambda_1 t}}{\lambda_1 - \lambda_2} = \frac{-1 - 0}{-1} = 1,$$

$$e^{\boldsymbol{A}t} = \boldsymbol{I} + (1 - e^{-t})\boldsymbol{A} = \begin{bmatrix} 1 - (1 - e^{-t}) & 0 \\ 1 - e^{-t} & 1 \end{bmatrix} = \begin{bmatrix} e^{-t} & 0 \\ 1 - e^{-t} & 1 \end{bmatrix}.$$

Check: $t = 0$ gives $\boldsymbol{I}$ ✓; derivative at 0 gives $\begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} = \boldsymbol{A}$ ✓. (Directly: $x_1 = e^{-t}x_{1,0}$, $x_2 = x_{2,0} + (1 - e^{-t})x_{1,0}$.)

Step 3 — Exact discretization. $e^{\boldsymbol{A}\alpha}\boldsymbol{B}$ is the first column, $[e^{-\alpha},\ 1 - e^{-\alpha}]^\top$:

$$\boldsymbol{A}_d = \begin{bmatrix} e^{-T_s} & 0 \\ 1 - e^{-T_s} & 1 \end{bmatrix}, \qquad \boldsymbol{B}_d = \int_0^{T_s} \begin{bmatrix} e^{-\alpha} \\ 1 - e^{-\alpha} \end{bmatrix} \mathrm{d}\alpha = \begin{bmatrix} 1 - e^{-T_s} \\ T_s - 1 + e^{-T_s} \end{bmatrix}.$$

($\int_0^{T_s} e^{-\alpha}\mathrm{d}\alpha = [-e^{-\alpha}]_0^{T_s} = 1 - e^{-T_s}$; $\int_0^{T_s}(1 - e^{-\alpha})\mathrm{d}\alpha = T_s - (1 - e^{-T_s})$.)

$$\boxed{\boldsymbol{x}[k+1] = \begin{bmatrix} e^{-T_s} & 0 \\ 1 - e^{-T_s} & 1 \end{bmatrix} \boldsymbol{x}[k] + \begin{bmatrix} 1 - e^{-T_s} \\ T_s - 1 + e^{-T_s} \end{bmatrix} u[k], \qquad y[k] = \begin{bmatrix} 0 & 1 \end{bmatrix} \boldsymbol{x}[k]}$$

Step 4 — Forward Euler.

$$\boxed{\tilde{\boldsymbol{A}}_d = \begin{bmatrix} 1 - T_s & 0 \\ T_s & 1 \end{bmatrix}, \qquad \tilde{\boldsymbol{B}}_d = \begin{bmatrix} T_s \\ 0 \end{bmatrix}, \qquad \boldsymbol{C},\ \boldsymbol{D} \text{ unchanged}}$$

Comparison.

- *Consistency:* expanding the exact result for small T_s ($e^{-T_s} \approx 1 - T_s$) gives $\boldsymbol{A}_d \approx \tilde{\boldsymbol{A}}_d$ and $\boldsymbol{B}_d \approx [T_s,\ 0]^\top = \tilde{\boldsymbol{B}}_d$ (the second entry $T_s - 1 + e^{-T_s} \approx T_s^2/2$ is second order) ✓.
- *Eigenvalues:* exact e^{-T_s} and 1; Euler $1 - T_s$ and 1. The lag pole stays inside the unit circle for **every** T_s in the exact model, but for Euler only if $|1 - T_s| < 1$, i.e. $T_s < 2$ (Concepts §9.3). The eigenvalue 1 is the integrator, which both methods keep.

- *Euler's $\tilde{\boldsymbol{B}}_d$ has a zero in the second entry:* the input reaches the output x_2 only one step later, whereas the exact model already moves x_2 during the first interval.

Check: `python/task4_discretization.py` (symbolic, and against `scipy.signal.cont2discrete(..., method="zoh")` for $T_s = 0.1, 0.5, 2, 5$).

Task 5 — Discretization 2

$$\ddot{y}(t) + \dot{y}(t) = u(t).$$

5(a) — State-space model and discretization

State-space model ($x_1 = y$, $x_2 = \dot{y}$; $\ddot{y} = u - \dot{y}$):

$$\dot{\boldsymbol{x}} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}}_{\boldsymbol{A}} \boldsymbol{x} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\boldsymbol{B}} u, \qquad y = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\boldsymbol{C}} \boldsymbol{x}, \quad \boldsymbol{D} = 0.$$

(A mass with friction: velocity x_2 relaxes towards the force, position x_1 integrates the velocity.) Eigenvalues 0 and -1 (upper triangular) — $\boldsymbol{A}$ is **singular**.

$e^{\boldsymbol{A}t}$ **with Cayley–Hamilton** ($\lambda_1 = 0$, $\lambda_2 = -1$):

$$\alpha_1 = \frac{e^0 - e^{-t}}{0 - (-1)} = 1 - e^{-t}, \qquad \alpha_0 = \frac{0 \cdot e^{-t} - (-1) \cdot e^0}{1} = 1, \qquad e^{\boldsymbol{A}t} = \boldsymbol{I} + (1 - e^{-t})\boldsymbol{A} = \begin{bmatrix} 1 & 1 - e^{-t} \\ 0 & e^{-t} \end{bmatrix}.$$

Check: $t = 0 \Rightarrow \boldsymbol{I}$ ✓, derivative at $0 \Rightarrow \boldsymbol{A}$ ✓.

Exact discretization. $e^{\boldsymbol{A}\alpha}\boldsymbol{B}$ = second column $\begin{bmatrix} 1 - e^{-\alpha}, & e^{-\alpha} \end{bmatrix}^\top$:

$$\boxed{\boldsymbol{A}_d = \begin{bmatrix} 1 & 1 - e^{-T_s} \\ 0 & e^{-T_s} \end{bmatrix}, \qquad \boldsymbol{B}_d = \int_0^{T_s} \begin{bmatrix} 1 - e^{-\alpha} \\ e^{-\alpha} \end{bmatrix} \mathrm{d}\alpha = \begin{bmatrix} T_s - 1 + e^{-T_s} \\ 1 - e^{-T_s} \end{bmatrix}, \qquad \boldsymbol{C}_d = \begin{bmatrix} 1 & 0 \end{bmatrix}}$$

Forward Euler.

$$\boxed{\tilde{\boldsymbol{A}}_d = \boldsymbol{I} + T_s\boldsymbol{A} = \begin{bmatrix} 1 & T_s \\ 0 & 1 - T_s \end{bmatrix}, \qquad \tilde{\boldsymbol{B}}_d = \begin{bmatrix} 0 \\ T_s \end{bmatrix}, \qquad \boldsymbol{C}_d = \begin{bmatrix} 1 & 0 \end{bmatrix}}$$

5(b) — Step response for $T_s = 0.5$ and $T_s = 2\pi$

$u[k] = \sigma[k]$, $\boldsymbol{x}[0] = 0$. (The sheet says "sampling rate T_s"; T_s is the sampling *interval* in seconds.)

Closed form, exact model. Write $q = e^{-T_s}$.

- Velocity: $x_2[k+1] = q\,x_2[k] + (1 - q)$. Its fixed point is 1, so $x_2[k] = 1 - q^k$ (Concepts §6.3, scalar version).
- Position: $x_1[k+1] = x_1[k] + (1 - q)x_2[k] + (T_s - 1 + q)$. Summing from 0 to $k - 1$ with $x_2[i] = 1 - q^i$:

$$x_1[k] = \sum_{i=0}^{k-1} \left[(1-q)(1-q^i)+T_s-1+q\right] = \sum_{i=0}^{k-1} \left[T_s-(1-q)q^i\right] = kT_s-(1-q)\frac{1-q^k}{1-q} = kT_s-1+q^k.$$

$$\boxed{y[k] = kT_s - 1 + e^{-kT_s} \qquad \text{(exact)}}$$

This is **exactly** the continuous step response $y(t) = t - 1 + e^{-t}$ sampled at $t = kT_s$, for every T_s — as promised by exact discretization.

Closed form, Euler model. The same computation with q replaced by $p = 1 - T_s$ (now $\tilde{\boldsymbol{B}}_d = [0, T_s]^\top$): $x_2[k] = 1 - p^k$, $x_1[k+1] = x_1[k] + T_s x_2[k]$, so

$$x_1[k] = \sum_{i=0}^{k-1} T_s(1-p^i) = kT_s-T_s\frac{1-p^k}{T_s} = kT_s-1+p^k, \qquad \boxed{\tilde{y}[k] = kT_s - 1 + (1 - T_s)^k \qquad \text{(Euler)}}$$

The whole difference: Euler replaces e^{-T_s} by its first-order approximation $1 - T_s$.

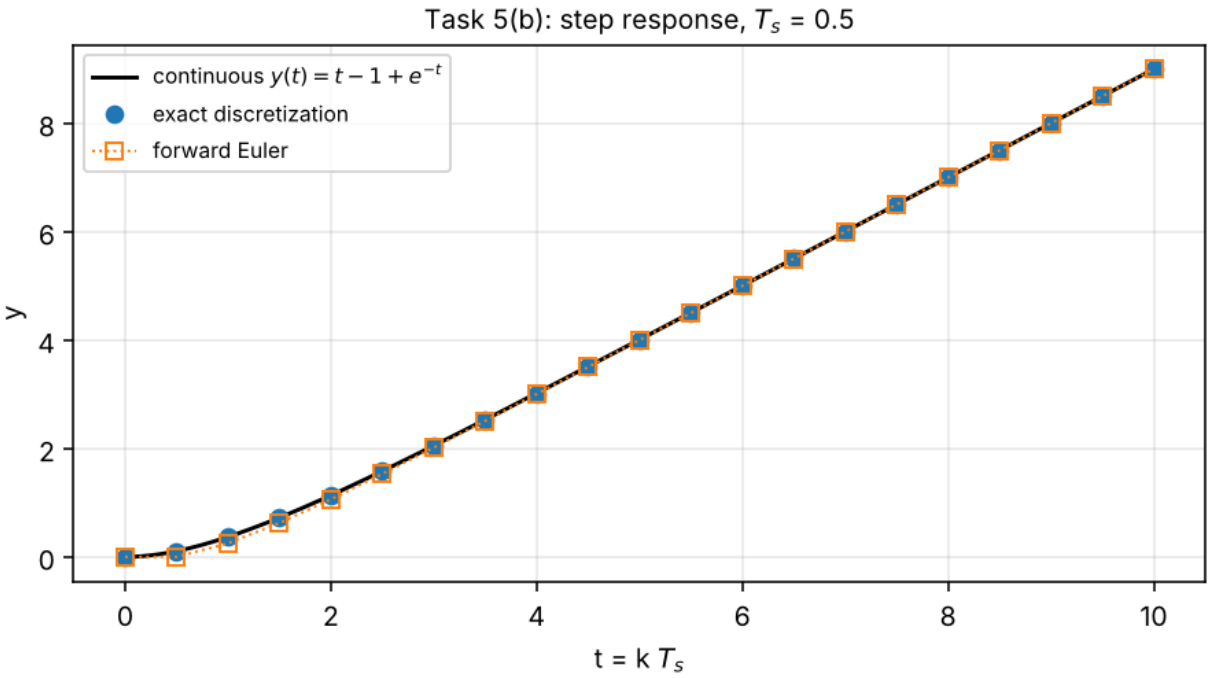
Values.

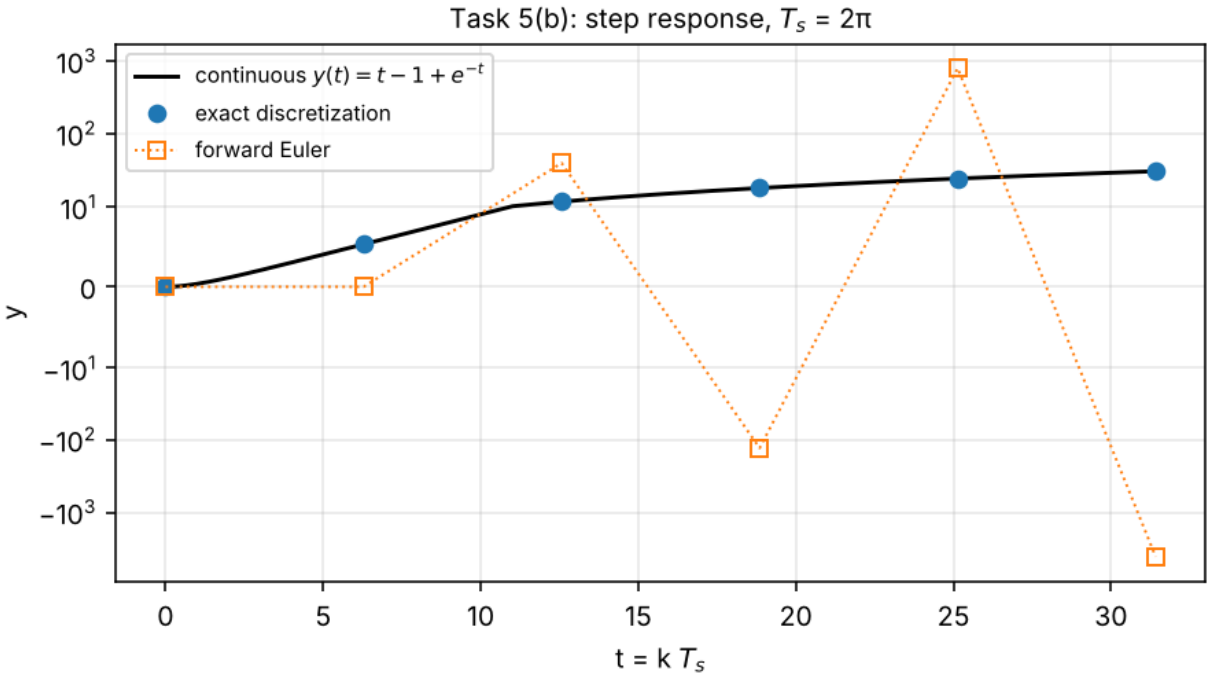
k	0	1	2	3	4	5	6
exact, $T_s = 0.5$	0	0.107	0.368	0.723	1.135	1.582	2.050
Euler, $T_s = 0.5$	0	0	0.250	0.625	1.063	1.531	2.016
exact, $T_s = 2\pi$	0	5.285	11.57	17.85	24.13	30.42	36.70
Euler, $T_s = 2\pi$	0	0	39.48	−129.6	803.2	−4086	21782

Discussion.

- $T_s = 0.5$**:** $e^{-0.5} = 0.607$ vs. $1 - 0.5 = 0.5$. Both models are stable (eigenvalues $\{1, 0.607\}$ and $\{1, 0.5\}$) and Euler stays close to the true curve; the error decays and the ramps become parallel. Euler always gives $\tilde{y}[1] = 0$, because the input reaches the position only through the velocity one step later.
- $T_s = 2\pi \approx 6.28$**:** the exact model is still perfect at the samples ($e^{-2\pi} \approx 0.002$ — the velocity has practically settled within one interval). Euler's eigenvalue is $1 - 2\pi \approx -5.28$, far outside the unit circle: the Euler model is **unstable** and oscillates with exploding amplitude, although the continuous system is perfectly well-behaved. The stability bound is $T_s < 2$ (Concepts §9.3, pole -1).
- The *continuous* plant itself is not asymptotically stable (eigenvalue 0: the position keeps growing like $t - 1$ under a constant force), but that growth is linear and bounded in slope; Euler's divergence for $T_s = 2\pi$ is exponential and purely a numerical artefact.

Check: `python/task5_discretization.py` (closed forms against simulation for both T_s).





Summary of results

- **Task 1:** $\boldsymbol{x}[1] = [0,0]^\top$, $\boldsymbol{x}[2] = [1,1]^\top$, $\boldsymbol{x}[3] = [2,0]^\top$; $\boldsymbol{x}[k] = [1 - \sin\frac{k\pi}{2},\ -\cos\frac{k\pi}{2}]^\top$; $y[k] = 1 - \sin\frac{k\pi}{2}$ (period 4: $1, 0, 1, 2, \ldots$).
- **Task 2:** Cayley–Hamilton ⇒ $\boldsymbol{A}^m\boldsymbol{B} \in \text{span}\{\boldsymbol{B}, \boldsymbol{AB}\}$, so 2 steps suffice; $u[0] = \frac{12}{13}$, $u[1] =$

$\frac{28}{13}$; general n: n steps; nonzero $\boldsymbol{x}[0]$: **not** in general (counterexample $\boldsymbol{A} = \text{diag}(2,1)$, $\boldsymbol{B} = [0,1]^\top$), but yes if $\text{rank}[\boldsymbol{B} \ \cdots \ \boldsymbol{A}^{n-1}\boldsymbol{B}] = n$.

- **Task 3:** both systems have $H[k] = (-1)^{k-1} - (-2)^{k-1}$. System 1: impulse $y = (-2)^{k-1}$ ($k \geq 1$, $y[0] = 0$), step $y = \frac{1}{6} + \frac{1}{2}(-1)^k - \frac{2}{3}(-2)^k$. System 2: impulse $y = -(-2)^{k-1}$ ($k \geq 1$, $y[0] = 1$), step $y = \frac{1}{6} + \frac{1}{2}(-1)^k + \frac{1}{3}(-2)^k$.
- **Task 4:** (a) $\boldsymbol{A}_d = \tilde{\boldsymbol{A}}_d = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}$, $\boldsymbol{B}_d = [\frac{T_s^2}{2}, T_s]^\top$, $\tilde{\boldsymbol{B}}_d = [0, T_s]^\top$. (b) $\boldsymbol{A}_d = \begin{bmatrix} e^{-T_s} & 0 \\ 1 - e^{-T_s} & 1 \end{bmatrix}$, $\boldsymbol{B}_d = [1 - e^{-T_s},\ T_s - 1 + e^{-T_s}]^\top$; $\tilde{\boldsymbol{A}}_d = \begin{bmatrix} 1 - T_s & 0 \\ T_s & 1 \end{bmatrix}$, $\tilde{\boldsymbol{B}}_d = [T_s, 0]^\top$.
- **Task 5:** $\boldsymbol{A}_d = \begin{bmatrix} 1 & 1 - e^{-T_s} \\ 0 & e^{-T_s} \end{bmatrix}$, $\boldsymbol{B}_d = [T_s - 1 + e^{-T_s},\ 1 - e^{-T_s}]^\top$; $\tilde{\boldsymbol{A}}_d = \begin{bmatrix} 1 & T_s \\ 0 & 1 - T_s \end{bmatrix}$, $\tilde{\boldsymbol{B}}_d = [0, T_s]^\top$; step responses $y[k] = kT_s - 1 + e^{-kT_s}$ (exact) and $kT_s - 1 + (1 - T_s)^k$ (Euler) — Euler unstable for $T_s = 2\pi$.